

Module 2

Supporting Cabling and Physical Installations

Learning Objectives

- Summarize Ethernet standards.
- Summarize copper cabling and connector types.
- Summarize fiber optic cabling and connector types.
- Describe physical installation factors for rack-based installations in server rooms and datacenters.
- Deploy and troubleshoot Ethernet cabling.

Lesson 2.1

Ethernet

Network Data Transmission Terms

Data transmission is transferring data from one device to another.

Data is sent point-to-point via wired or wireless streams or **channels**.

Digital signals (1s and 0s) are transmitted across the channels.

Transmission media include copper wires, fiber optic cables, or wireless signals.

The speed at which the data is sent is the **data transfer rate**.

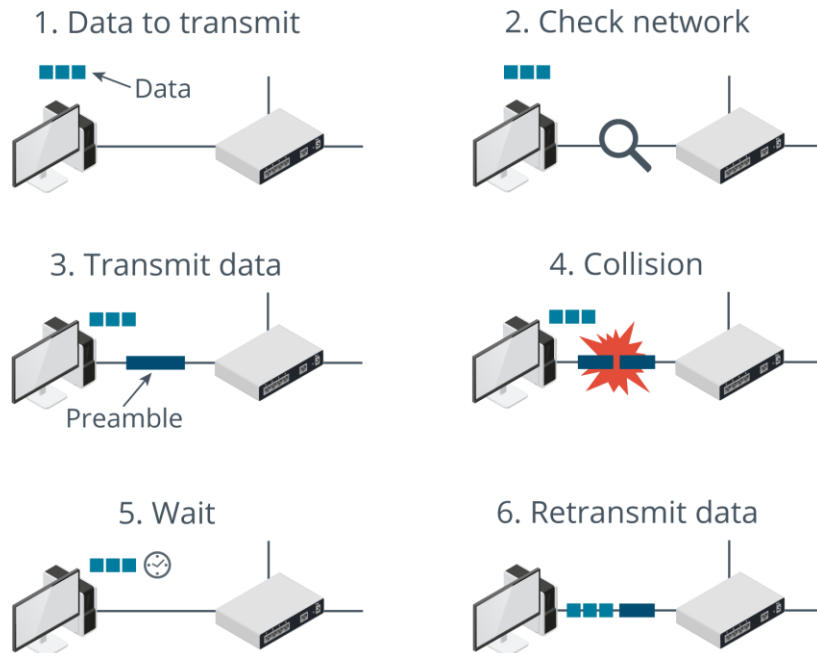
Error detection/correction mechanisms detect transmission errors.

Ethernet Standards

IEEE 802.3 standards define the physical layer and data link layer's media access control (MAC) for wired Ethernet.

Standard	Cables	Speed
10Base-T	Unshielded twisted pair (UTP)	10 Mbps
100Base-T	Cat5e or higher	100 Mbps
1000Base-T Gigabit Ethernet	Cat6 or higher	1000 Mbps
10GBase-T 10 Gigabit Ethernet	Cat6 or higher	10 Gbps
100Base-TX Fast Ethernet	Cat5 or higher	100 Mbps

CSMA/CD Diagram



Fiber Ethernet Standards

Standard	Cables	Speed
100Base-FX	MMF (OM1)	100 Mbps
100Base-SX	MMF (OM1, OM2)	100 Mbps
1000Base-SX	MMF (OM2, OM3)	1 Gbps
1000Base-LX	MMF (OM1, OM2, OM3), SMF (OS1, OS2)	1 Gbps
10GBase-SR	MMF (OM2, OM3, OM4)	10 Gbps
10GBase-LR	SMF (OS1, OS2)	10 Gbps

Activity: True or False

CSMA/CD regulates communication in networks with shared transmission media.



Activity: True or False

CSMA/CD regulates communication in networks with shared transmission media.



Lesson 2.2

Copper Cables and Connectors

Unshielded Twisted Pair (UTP)

Structure

- Contains 2 to 1800 pairs in a plastic jacket

Interference

- Minimal EMI protection

Use

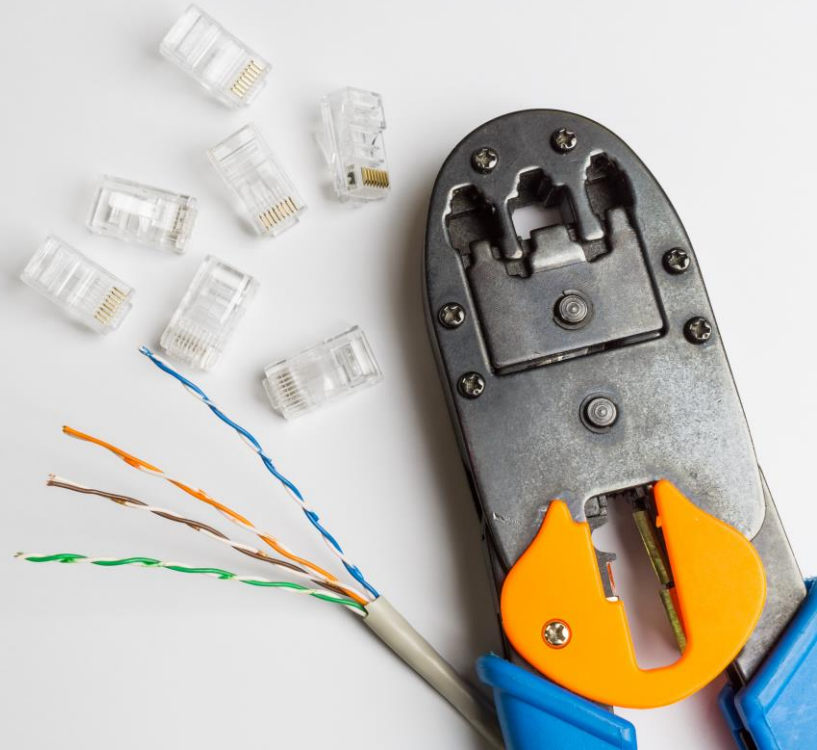
- Popular for LANs because it is cost-effective

Categories

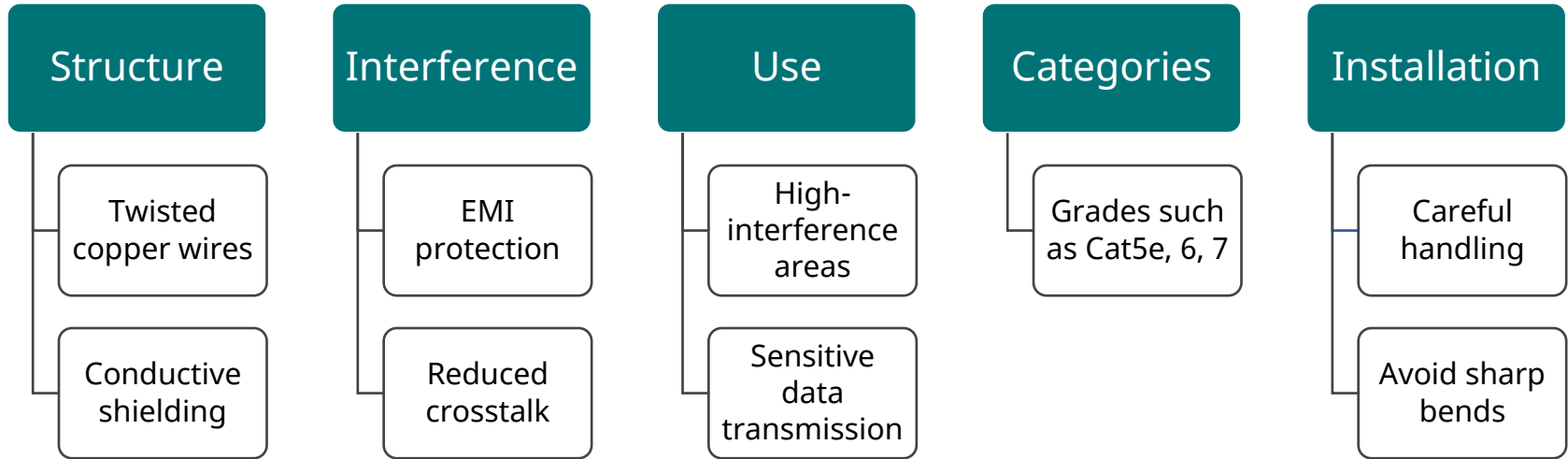
- Grades like Cat5e (1 Gbps) and Cat6a (10 Gbps)

Installation

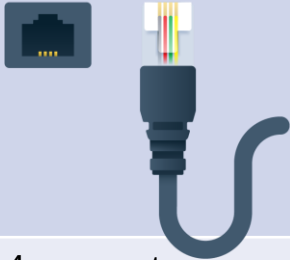
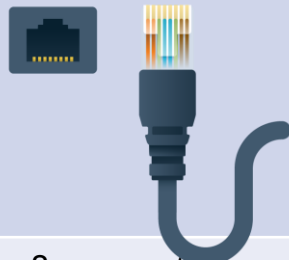
- Avoid sharp bends, keep away from EMI sources



Shielded Twisted Pair Cable (STP)



Ethernet Connectors

Feature	RJ11	RJ45
Image	 An illustration of an RJ11 connector, which is a small, dark grey plastic plug with four gold-colored pins. Next to it is a cable with a similar plug, showing four colored wires (yellow, green, blue, and red) inside the clear plastic sheath.	 An illustration of an RJ45 connector, which is a small, dark grey plastic plug with eight gold-colored pins. Next to it is a cable with a similar plug, showing eight colored wires (yellow, green, blue, red, orange, brown, white, and black) inside the clear plastic sheath.
Configuration	6 positions, 4 connectors	8 positions, 8 connectors
Usage	Phone and modem	Computer networking
Bandwidth	24 Mbps	10 Gbps over Ethernet

Coaxial and Twinaxial Cable

Feature	Coaxial	Twinaxial
Image		
Configuration	Central conductor, insulation, metallic shield, outer jacket	Two inner conductors in a twisted pair, insulation, outer jacket
Usage	TV, Internet, radio signals, CCTV	10 GB Ethernet networks
Bandwidth	Wide range of frequencies, high-speed data transmission	Very-short range high-speed
Connectors	BNC, TNC, SMA	Proprietary

Activity: Multiple Choice



A company is setting up a network in an industrial environment where machinery often causes significant electromagnetic interference. The network requires a cabling solution that can handle high-speed data transfer while also being resistant to this interference. The cable will be used to connect servers within the same data center, and the runs will not exceed 100 meters.

- A. Unshielded Twisted Pair (UTP)
- B. Shielded Twisted Pair (STP)
- C. Coaxial Cable
- D. Fiber Optic Cable

Activity: Multiple Choice



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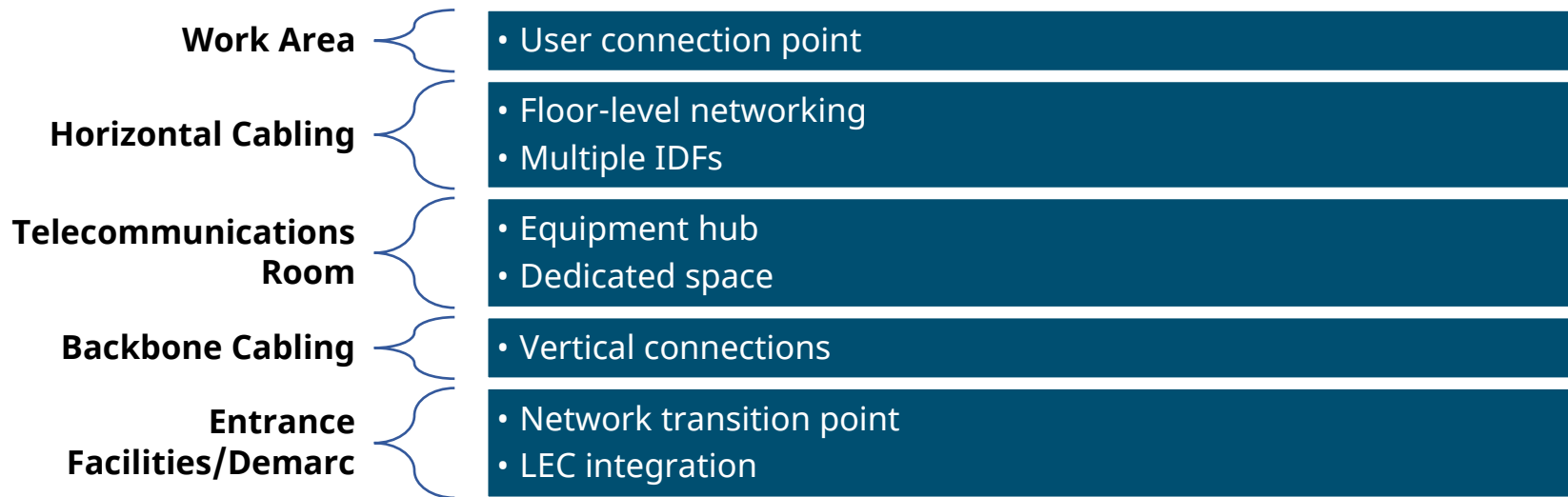
- A. Unshielded Twisted Pair (UTP)
- B. **Shielded Twisted Pair (STP)**
- C. Coaxial Cable
- D. Fiber Optic Cable

Lesson 2.3

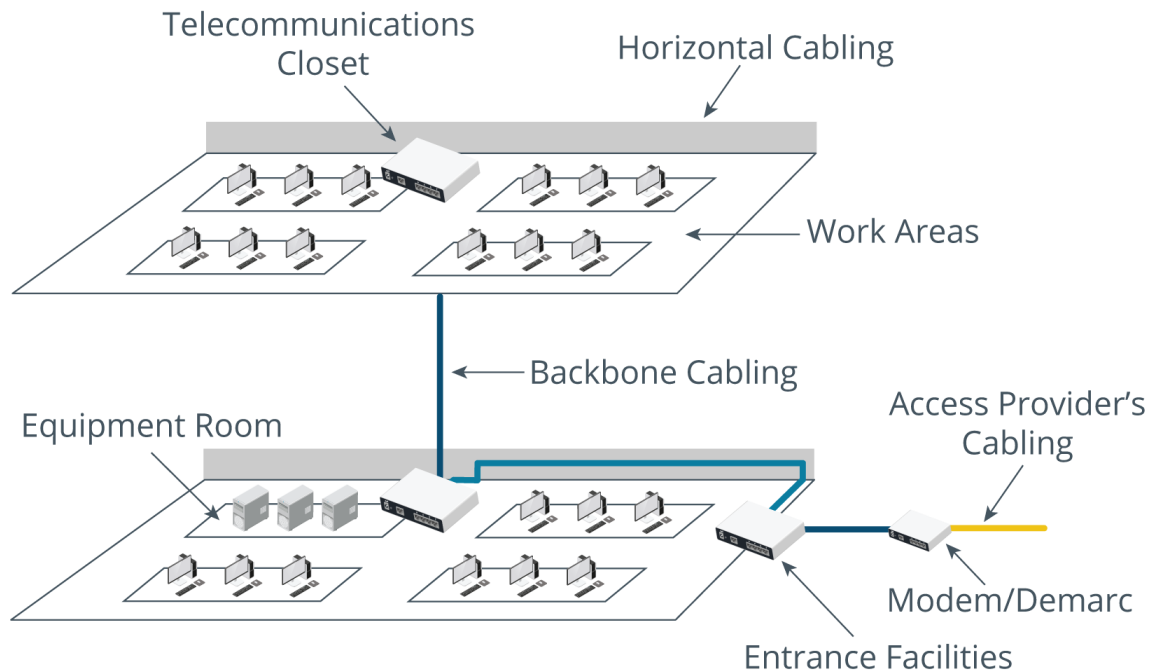
Wiring Implementation

Structured Cabling System

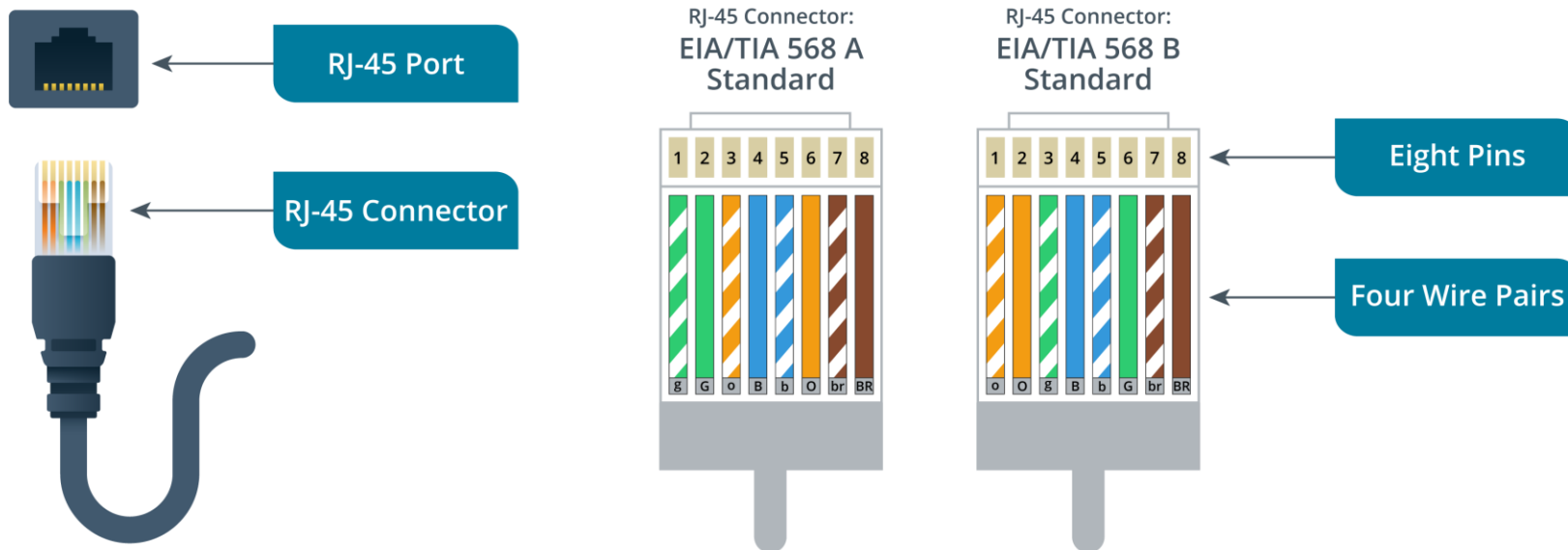
A structured cabling scheme is a standard way of provisioning cabled networking for computers in an office building.



Structured Cabling Diagram



T568A and T568B Termination Standards



Patch Panels

Purpose

- Facilitates cable management

Structure

- Punch down blocks at rear for cable termination
- RJ45 ports in front for device connections

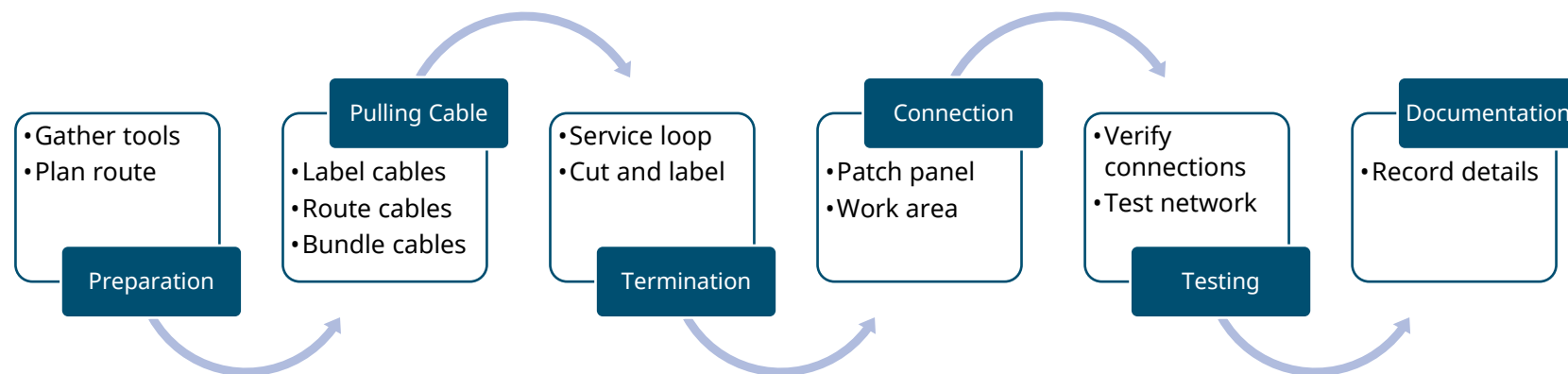
Function

- IDC terminals allow for secure and compact cable termination
- RJ45 ports enable easy reconfiguration of network connections

Wiring Standards

- Supports T568A wiring schemes
- Supports T568B wiring schemes

Structured Cable Installation



Termination Tools

Cable Cutters



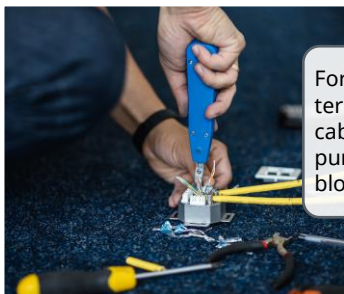
For clean cuts of the cable without damaging wires.

Crimp Tools



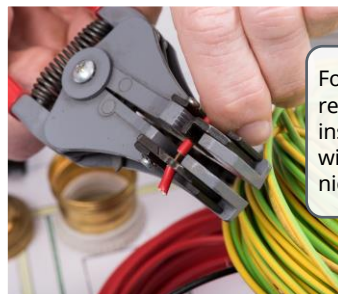
To attach connectors to the cable ends.

Punch-Down Tools



For terminating cable wires in punch down blocks.

Cable Strippers



For safely removing insulation without nicking wires.

Activity: What is it?

1



2



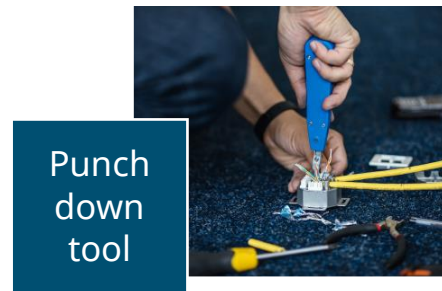
3



4



Activity: What is it?



Lesson 2.4

Fiber Optic Cables and Connectors

Fiber Optic Considerations

Fiber optic cables

- Thin strands of glass or plastic
- Transmit data using light pulses

Composition

- Core: ultra-pure strands of glass
- Cladding: reflects light back into the core
- Buffer: outer protective jacket

Benefits

- Transmit data up to 800 Gbps
- Maintain signal quality over long distances
- Immune to electromagnetic interference

Single Mode and Multimode Fiber

Factor	Single Mode	Multimode
Core diameter	Small core allowing only one light to pass through	Larger core allowing multiple light modes to pass through
Light transmission	Light travels directly down the fiber with minimal reflection and attenuation	Light bounces more causing more dispersion and attenuation
Distance and bandwidth	Suitable for long distances	Best for shorter distances
Applications	Long-haul networks, high-speed broadband, telecommunications	Datacenters, LANs
Cost	More expensive	Less expensive

Fiber Optic Connector Types

Factor	Subscriber Connector (SC)	Lucent Connector (LC)	Straight Tip Connector (ST)
Image			
Ferrule Size	2.5 mm	1.25 mm	2.5 mm
Applications	Telecommunications networks, data transmission	Heavily populated patch panels and cabinets	LANs and datacenters
Features	Push-pull design, quick deployment, low insertion loss, durability	Square shape, duplex header design	Bayonet twist-lock, older but still widely used

Wavelength Division Multiplexing (WDM)

Allows multiple data channels to share the same fiber using different wavelengths of light

Bidirectional WDM (BiDi):

- Supports Tx and Rx over the same fiber strand
- Uses shifted wavelengths (1310 nm for Tx, 1490 nm for Rx)
- Requires installation in opposite pairs
- Documented in Ethernet standards (1000BASE-BX, 10GBASE-BX)

Coarse WDM (CWDM):

- Supports up to 16 wavelengths
- Typically used for 4-8 bidirectional channels on a single strand
- Can also be used for unidirectional channels on dual strands

Dense WDM (DWDM):

- Provisions more channels (20, 40, 80, or 160)
- Requires precise lasers due to less spacing between channels
- Supports multi-channel 1G, 10G, and 40G Ethernet links
- Transceivers installed in opposite pairs for proper function

Activity: Two Truths and a Lie

Fiber optic cables are made of thin strands of glass or plastic that transmit data using light pulses

Single mode fiber causes more dispersion and attenuation than multimode fiber

Lucent connectors are often used for heavily populated patch panels



Activity: Two Truths and a Lie

Fiber optic cables are made of thin strands of glass or plastic that transmit data using light pulses

Single mode fiber causes more dispersion and attenuation than multimode fiber

LIE

Lucent connectors are often used for heavily populated patch panels



Lesson 2.5

Physical Installation Factors

Rack Systems



Secure Areas:

- Telecommunications closets, equipment rooms, server rooms
- Datacenters for server provisioning

Access Control:

- Physical access controls for authorized entry

Rack Installation:

- Steel shelving systems for standard-size equipment
- Secure, compact storage; increased density

Standard Sizing:

- EIA standard 19"/ 48.26 cm width for rack-mounting

Humidity and Temperature

Temperature Monitoring

Prevents overheating

Ensures effective heat dissipation

Humidity Control

Avoids condensation and corrosion

Reduces static charge risks

Electrical Stability

Ensures uninterrupted power supply

Monitors for voltage irregularities

Flood Detection

Identifies water-related risks

Triggers immediate system shutdowns



Power Management

Power Essentials

Essential stable power for network appliances

Protection against voltage spikes, surges, and failures

Load and Voltage

AC circuits aligned with equipment load

High-voltage circuits in datacenters

PDUs

Ensure clean power signals

Offer remote monitoring/control

Battery Backups

Component-level storage device protection

UPS for system-level power continuity

UPS runtime dependent on load/model

Fire Suppression

Fire Safety Elements

Clearly marked fire exits.

Regular emergency evacuation drills.

Fire-resistant building design.

Detection and Alarms

Automatic and manual smoke/fire detection systems.

Extinguishers:

Various types for different fire classes.

Class C extinguishers for electrical safety.

Sprinkler Systems

Heat-triggered, water discharge.

Risk of burst pipes, accidental triggering.

Sprinkler Alternatives

Dry-pipe

Pre-action

Halon (phased out).

Clean agent

Lesson 2.6

Cable Troubleshooting

Specifications and Limitations: Part One

Understanding Specifications

- Compare expected vs. actual performance
- Assess speed, throughput, distance

Speed vs. Throughput

- Physical Layer: Symbols transmitted, measured in baud rate (Hz)
- Data Link Layer: Nominal bit rate or bandwidth (bps)

Throughput Factors

- Average data transfer rate over time
- Affected by encoding, errors, distance, interference

Specifications and Limitations: Part Two

Measurement Layers

- Network/Transport Layer: Throughput
- Application Layer: Goodput (accounting for packet loss)

Latency

- Speed of packet delivery, measured in milliseconds (ms)
- Also known as latency or delay

Distance Limitations

- Media type dictates bit rate over distance
- Attenuation (dB loss) and interference (SNR) impact performance

Cable Issues: Symptoms

Slow Internet
speeds

Random
disconnections
and
reconnections

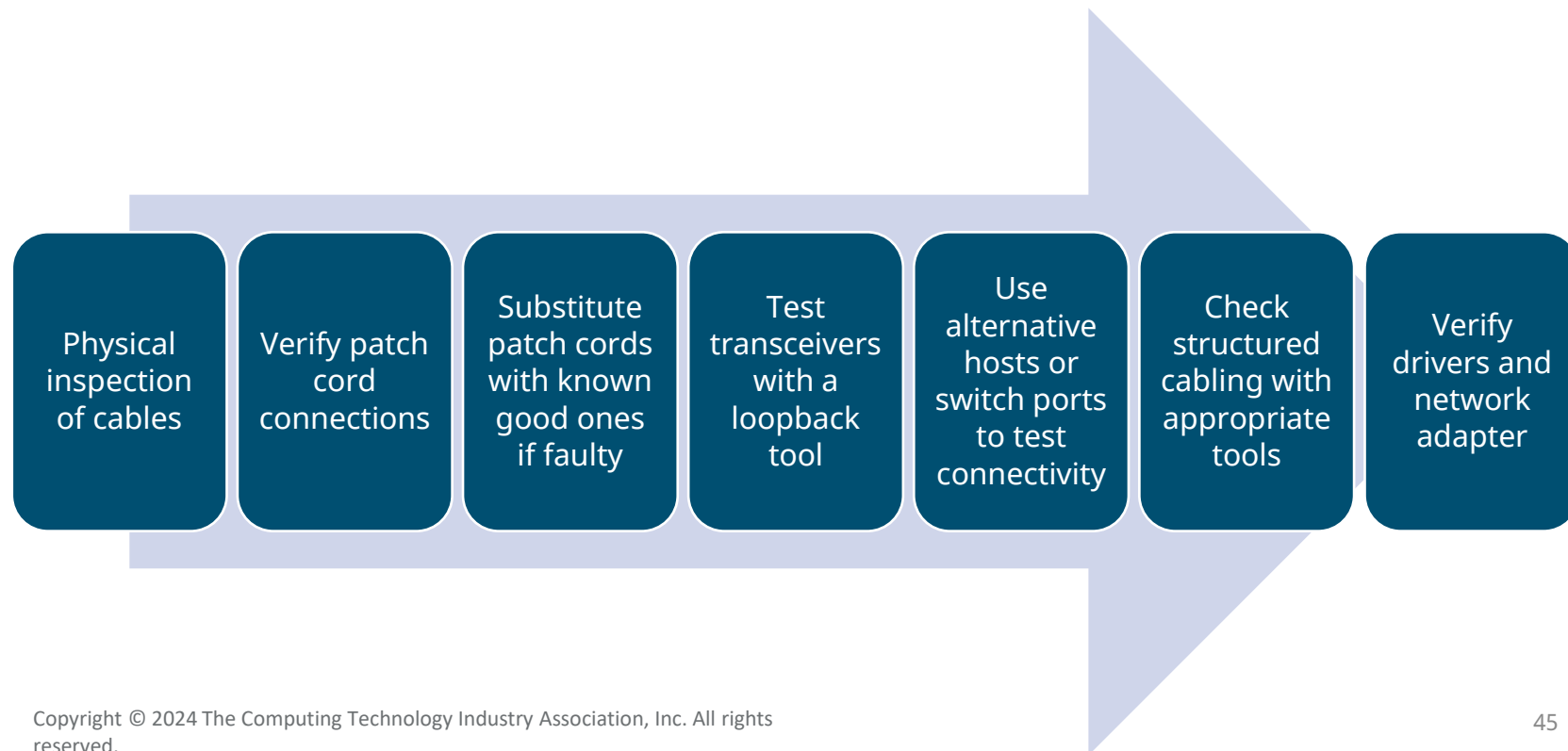
Connection
timeouts

Slowing down
of applications

Lagging
audio/video
communication

Frequent drops
or unstable
network link

Cable Issues: Troubleshooting



Cable Testers

Diagnosis with Cable Testing Tools

- Used when cable is not directly accessible
- Diagnose intermittent connectivity or performance issues

Cable Tester Functions

- Reports on physical and electrical properties
- Tests conditions, crosstalk, attenuation, noise, resistance



Wire Map Testers and Tone Generators

Wire Map Testers

- Detect improper cable termination
- Use base and remote units to test each wire conductor
- Identify issues like open circuits, shorts, and incorrect pin-outs

Tone Generators

- Trace cables through walls or identify active cables in a bundle
- Known as “Fox and Hound”
- Apply signal to trace cable with a probe

Attenuation Issues

Loss of signal strength in networking cables or connections
(measured in decibels (dB) or voltage)

Issues

- Signal strength loss during transmission
- Higher impedance, higher attenuation
- Long cabling distance
- Thin wire size
- Environmental factors

Solutions

- Shorten cable lengths
- Use repeaters/extenders
- Upgrade to high-quality cables
- Measure and test signal attenuation at installation

Interference Issues

Negative effects of electromagnetic, radio frequency, and electrostatic signals on cable transmissions

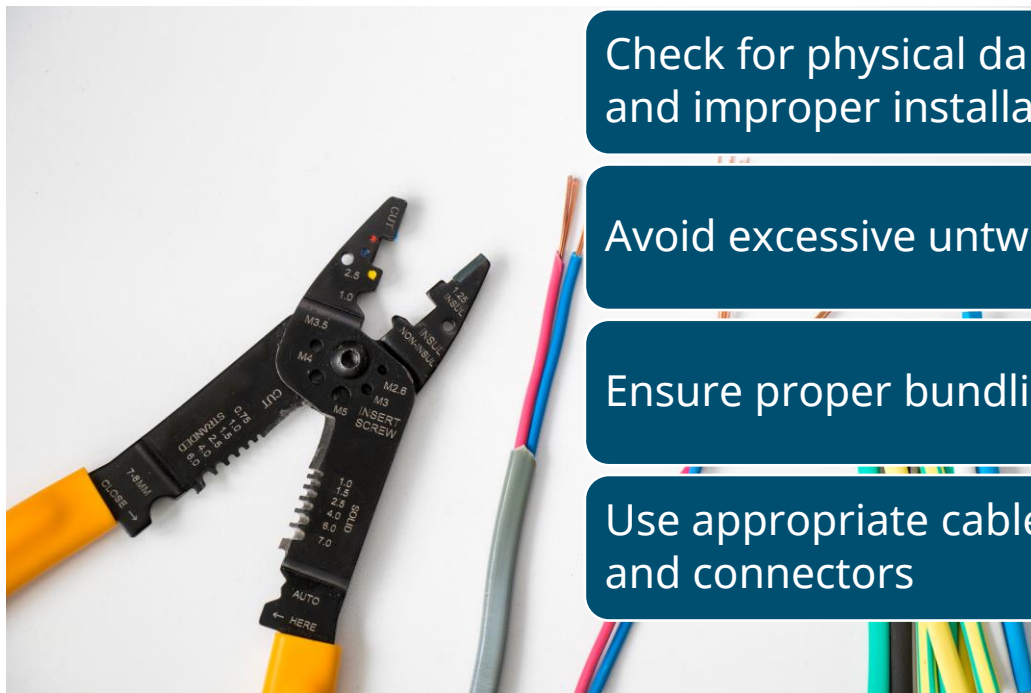
Issues

- Electromagnetic interference (EMI)
- Radio frequency interference (RFI)
- Crosstalk from adjacent wires
- Defective connectors/conductors

Solutions

- Shielding techniques (foil/braided shields)
- Systematic cable routing and organization
- Use robust materials and strain relief
- Compatibility testing

Crosstalk Issues: Solutions



Check for physical damage and improper installation

Avoid excessive untwisting

Ensure proper bundling

Use appropriate cable types and connectors

Activity: Think About It

What are some symptoms of cabling issues?



Summary

Understand Cable Needs: Consider factors like interference and attenuation when choosing shielded copper or fiber optic cables

Proper Cable Preparation: Use appropriate tools for cable preparation and termination for connectors or punch down blocks

Application-Specific Use: Ensure cables are used for their intended purpose to meet network requirements.

Cable Testing: Verify cable integrity using appropriate testing tools to identify and troubleshoot faults